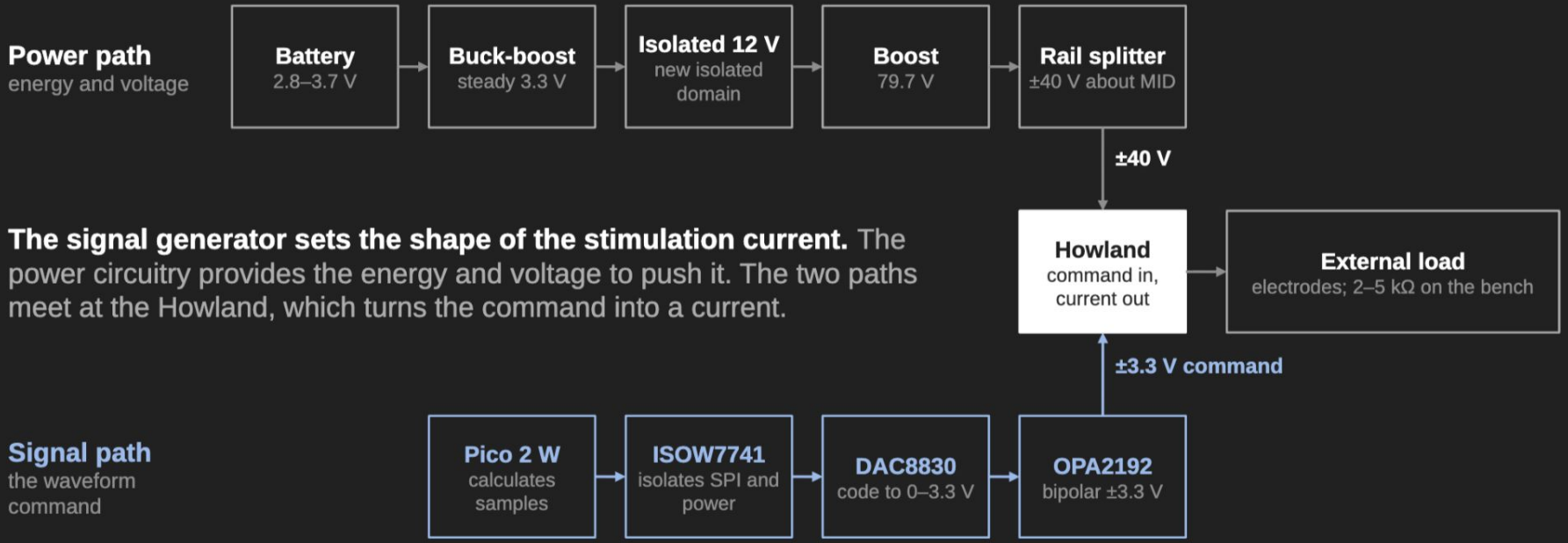
Prototype Demo 2

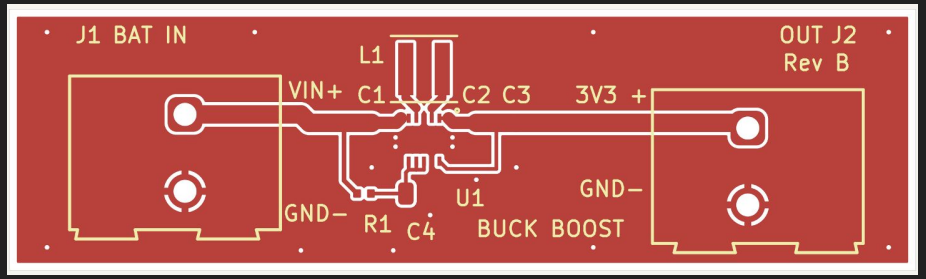
By Pythagoras Carayannopoulos, Peter Zhao, and Anish Lanka

How the hardware fits together



The signal generator sets the shape of the stimulation current. The power circuitry provides the energy and voltage to push it. The two paths meet at the Howland, which turns the command into a current.

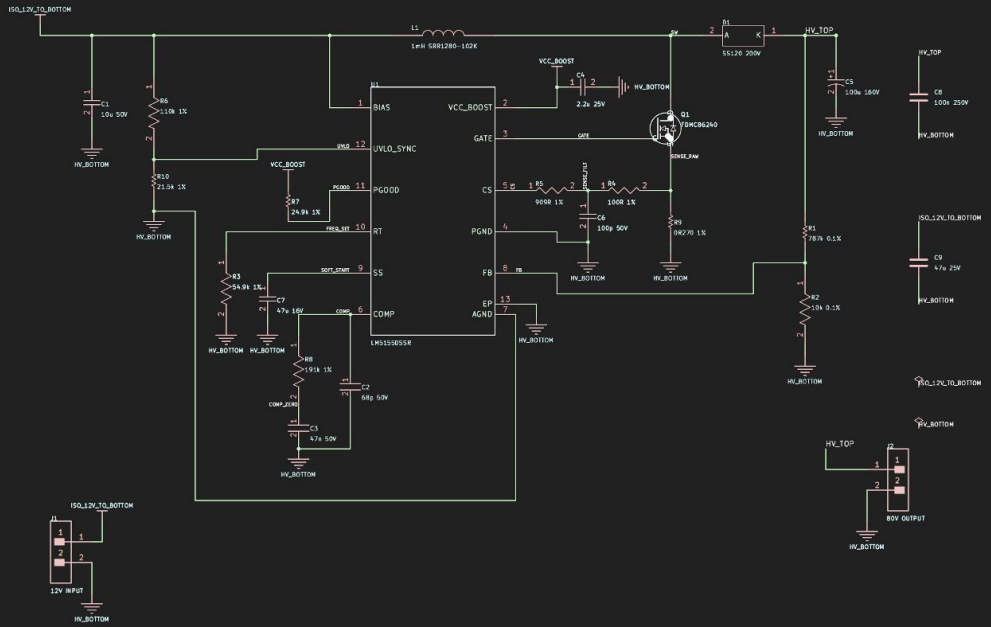
Buck-Boost



- Takes battery voltage as input.
  - Should be around 2.8V-3.7V (We don’t have our battery yet)
  - The reason for the range is that it is expected to start towards the higher end (3.7V) and gradually decrease to the lower end of the range (2.8V)
- Our Buck-Boost PCB takes this input voltage and steps it up or down to 3.3V

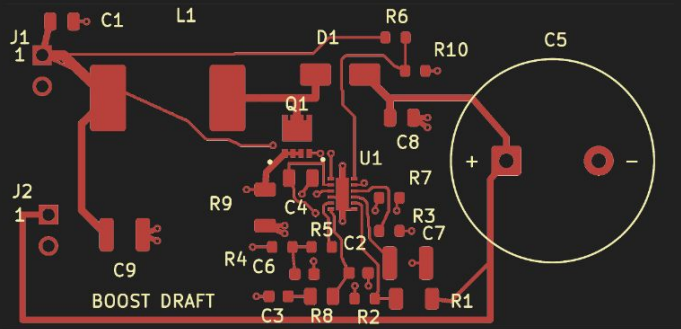


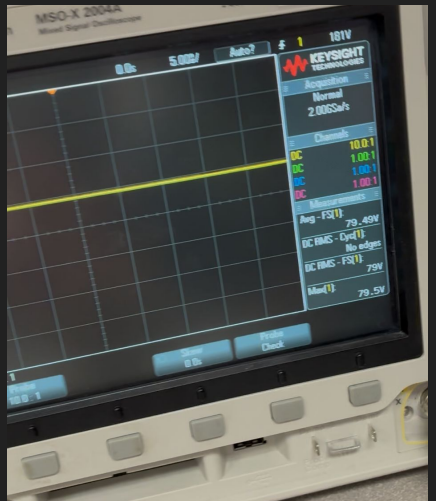
Boost converter



- Takes in 12V and steps it up to about 80 V
- Tested: 12 V in, 80 V out

Boost converter





- New feedback resistors set the output to 79.7 V: R1 = 787 kΩ and R2 = 10 kΩ
- Parts that see 80 V needed higher ratings:
  - MOSFET Q1: 100 V → 150 V
  - Diode D1: 100 V → 200 V
- Top feedback resistor: small 0603 (about 75 V max) → bigger 1206 (200 V)
- Output capacitor C5: 22 µF at 100 V → 100 µF at 160 V
- Boost formula: Vout = Vin ÷ (1 − D). Going from 12 V to 80 V, the switch is on about 85% of the time

What we built on

STIM THE BRIM:

~60 V

Original boost output

Nominal split rails: +30 V and -30 V

OUR DESIGN:

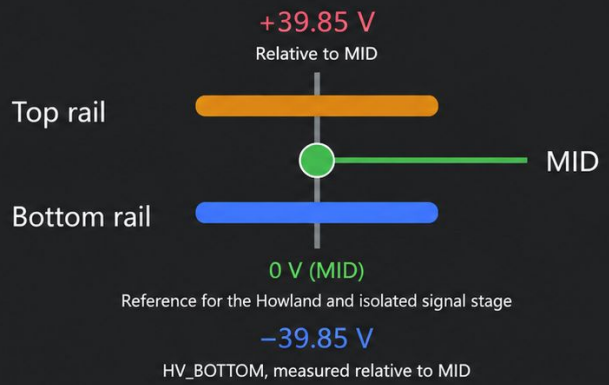
~80 V

Original boost output

Nominal split rails: +40 V and -40 V

We did 40 V so firmware can limit it to 30 V therefore no clipping due to rail saturation

Rail Splitter - Background

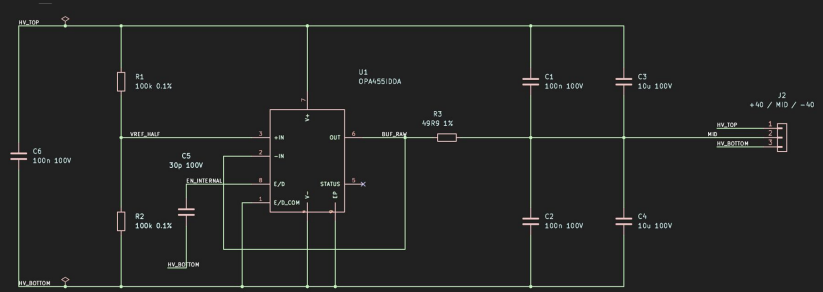


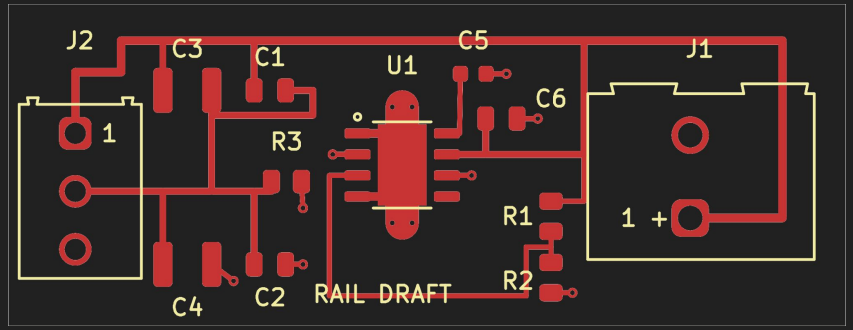
If you measure relative to the boost converter's bottom rail instead, the same three nodes would be approximately:

Node	Relative to HV_BOTTOM	Relative to MID
HV_TOP	79.7 V	+39.85 V
MID	39.85 V	0 V
HV_BOTTOM	0 V	−39.85 V

Rail Splitter

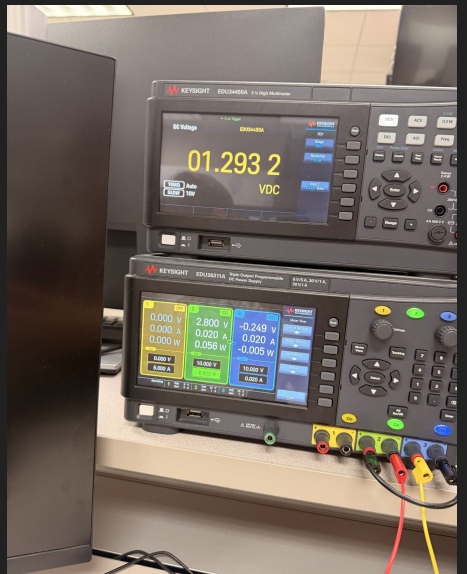
- Added extra capacitors to reduce noise and stabilize voltage.
- Added a resistor at the output to improve stability.
- Added capacitors across the power rails for filtering.
- Included a connector to easily connect the rail splitter to other PCBs.
- Used matched resistors to create an even voltage split.





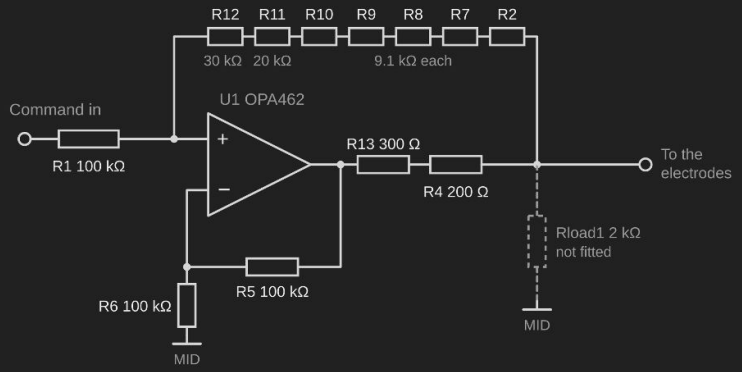
Rail Splitter

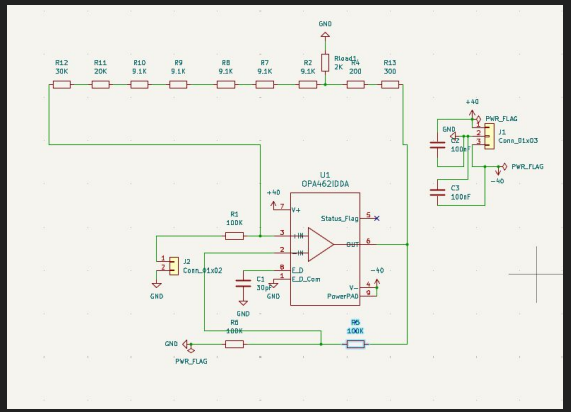
- During testing we were unable to get our voltage high
  - We believe the op-amp may have been damaged and could be causing the rail splitter to malfunction.
- Despite this, the circuit appeared to split the input voltage evenly, but further testing is needed to verify proper operation.

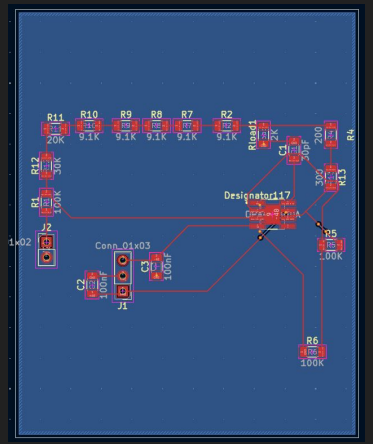


Howland Pump

- Theoretically: The Howland pump should be able to give up to +- 5mA with +-30V, but we made our rails +-40V because of saturation near the voltage limit instead of +-30V
- In Reality: When we tested our Howland, we got a peak-to-peak value 96V, which is perplexing and we need to investigate that this week



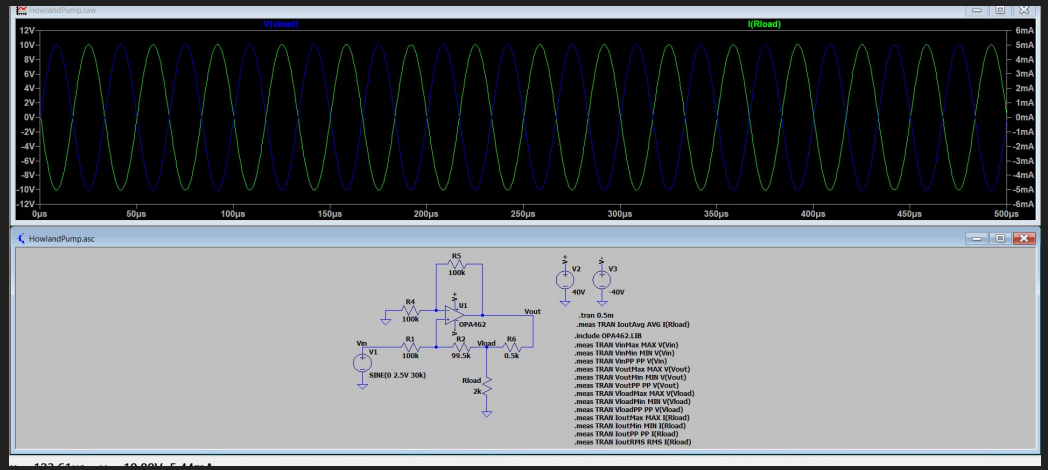




Howland Pump Testing

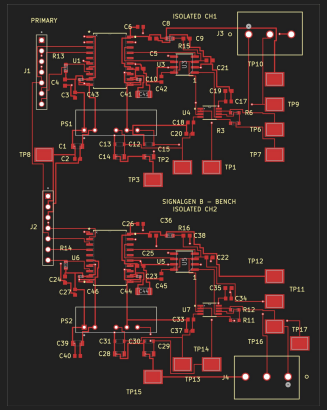
- For some unknown reason our howland pump did not work, but the LTspice simulation proved its functionality and we did not have time to investigate what went wrong.

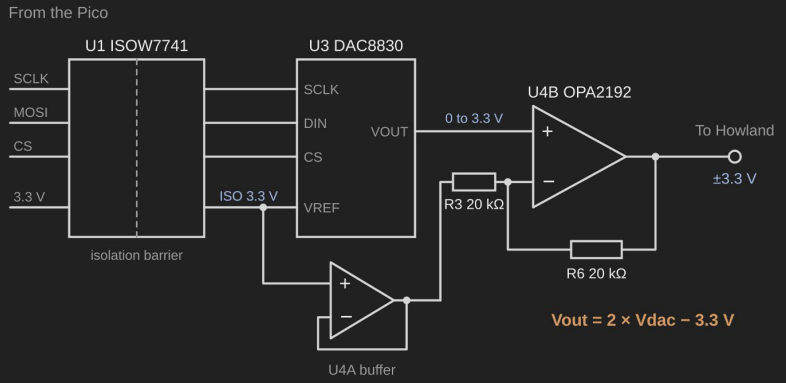




Signal Generator - The Signal Path

- Raspberry Pi Pico/Controller (calculates the waveform samples)
- ISOW7741 (isolates digital SPI signals and supplies isolated power)
- DAC8830 (converts 16-bit digital commands into analog voltage)
- OPA2192 (buffers the reference and creates a bipolar output)
- OUTPUT (approximately -3.3V to 3.3V) → Howland Pump





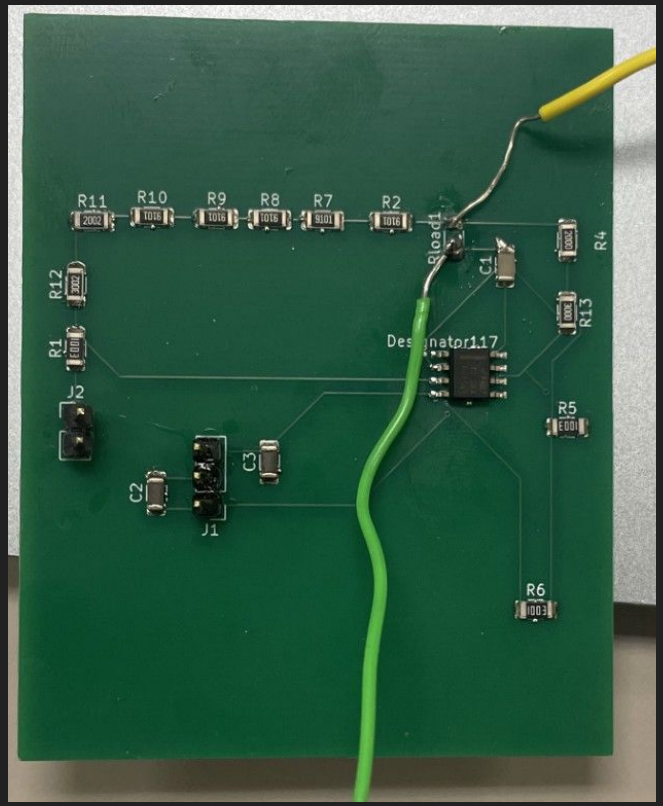
SigGen for 3.3V Reference We Get (Expected)

DAC output (V)	Final Bipolar Output (V)
0	-3.3
0.825	-1.65
1.65	0
2.475	1.65
3.3	3.3

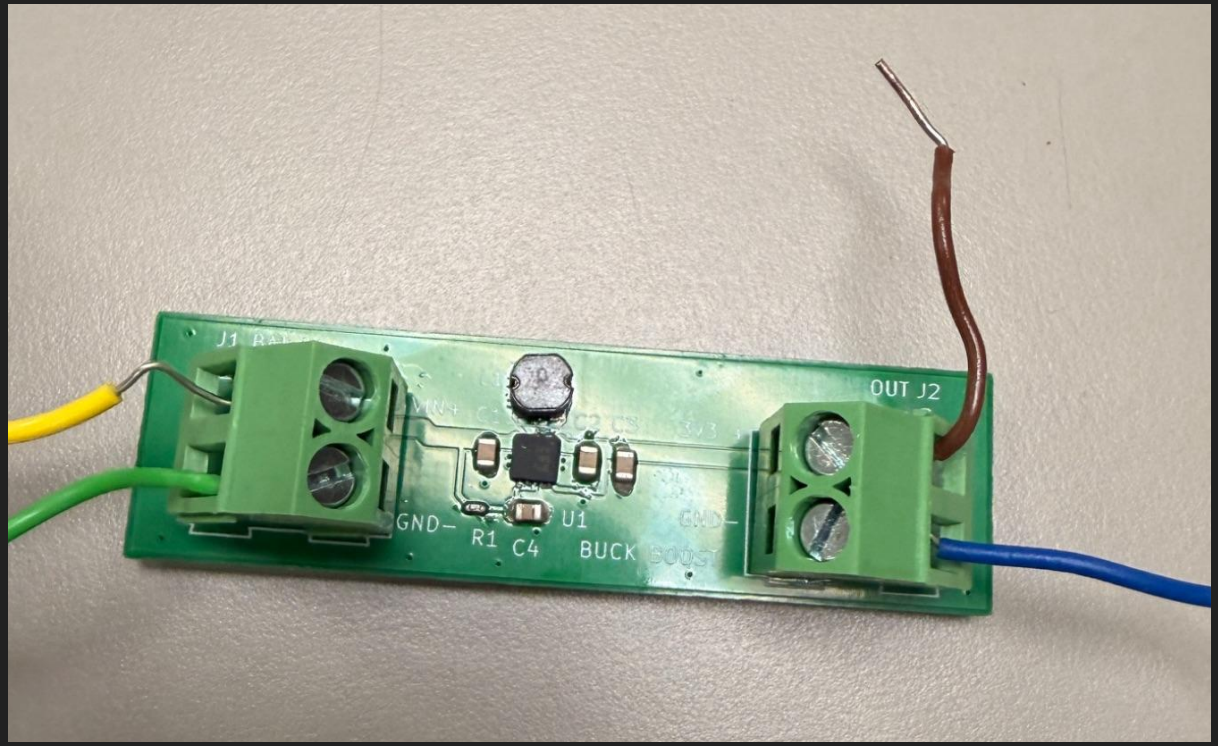
Signal Generator fixes

- The DAC output was wired straight into a 10 kΩ resistor. The DAC8830 can’t drive that (its output acts like a 6.25 kΩ resistor), so the voltage came out wrong. Now the OPA2192 buffers it
- The board had the footprint for the old 6-pin DAC8411, but the DAC8830 has 8 pins. We changed the footprint
- At power-up the DAC outputs 0, which comes out as −3.3 V (full negative). The firmware has to set it to the middle first
- Firmware: Modularization (Code organization, Typescript frontend web app, C backend, created Student view and Teacher view)

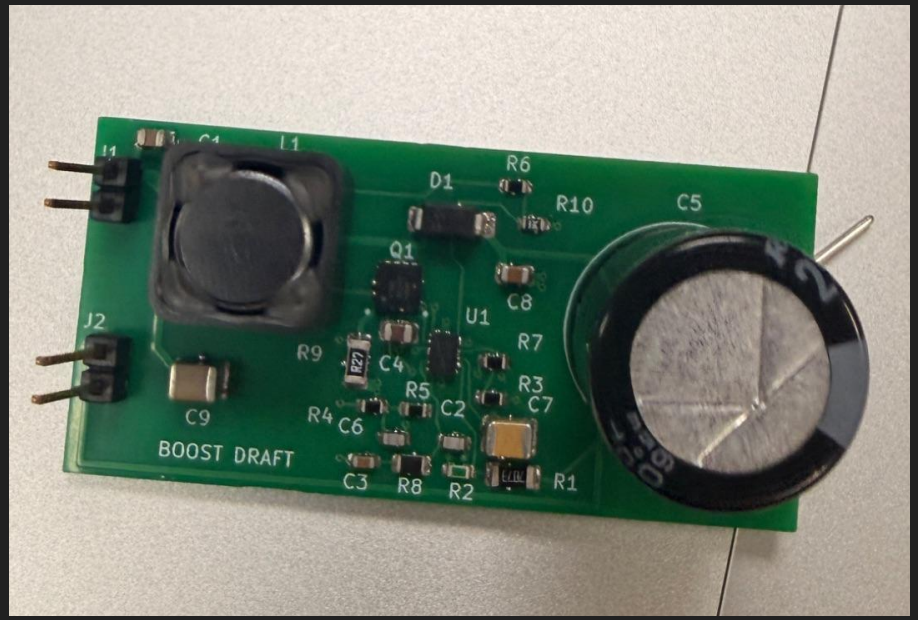
Soldered PCB - Howland pump



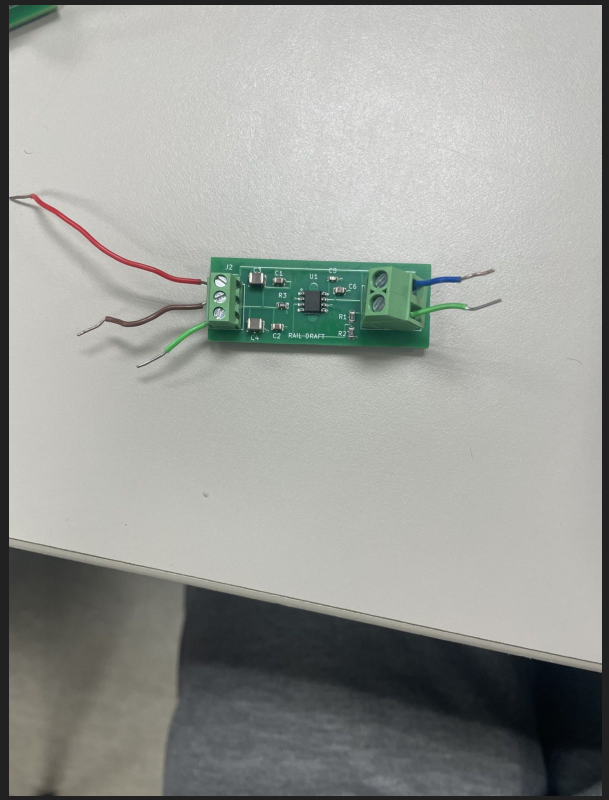
Soldered PCB - Buck Boost



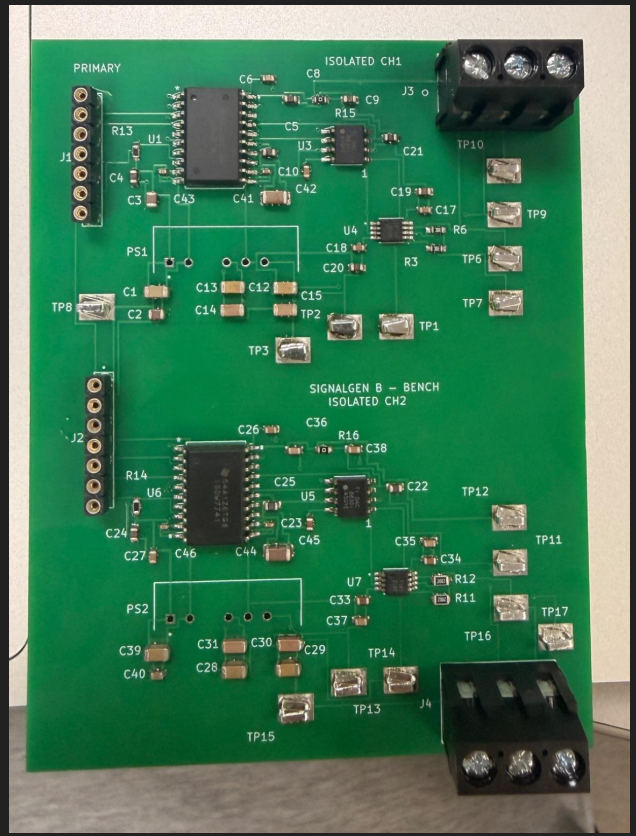
Soldered PCB - Boost Converter



Soldered PCB - Rail Splitter



Soldered PCB - Signal Generator



Who worked on what

Pyth → Boost, Howland Design, soldering, testing

Anish → Rail splitter, Buck-Boost, Howland PCB, soldering, testing

Peter → Signal Generator, initial firmware, testing